

Continuous Optimization for Airline Hub-and-Spoke Network Design

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Abstract—We study the design of airline hub-and-spoke networks through a continuous optimization framework that jointly models network deployment, passenger choice, and market profitability. The lower-level model combines operating costs, sparse deployment penalties, price-weighted passenger utility terms, and entropy regularizers that recover a logit demand split between the new airline and the outside alternative. This yields a continuous approximation of hub activation and route opening decisions while preserving the economic structure of the airline design problem.

On top of this model, we introduce a bilevel formulation in which market weights are learned rather than fixed. The upper level maximizes the real objective of the airline, namely profit, by adjusting market-specific hyperparameters that scale the lower-level utility and entropy terms. This allows the model to emphasize the origin-destination markets that are most relevant for profitability, instead of relying on exogenous weights chosen *a priori*.

The main difficulty is the simultaneous presence of hierarchical optimization and fixed deployment costs. To address it, we combine a reweighted ℓ_1 approximation of fixed costs with a penalty-based bilevel strategy. The resulting algorithm alternates an outer sparsity loop, which updates the reweighting coefficients associated with capacities, and an inner penalty loop, which alternates between a penalized bilevel solve and a lower-level solve to update the market hyperparameters. The final methodology provides a practical route to solving airline network design problems that are beyond the reach of exact mixed-integer bilevel formulations, while maintaining an explicit link between passenger behavior, network topology, and profit.

I. INTRODUCTION

Transportation networks are a critical component of modern societies. They support economic activity, territorial cohesion, access to employment and services, and the movement of passengers and goods at multiple spatial and temporal scales. However, the goals of network design depend strongly on the application. In road networks, the emphasis is typically placed on reducing travel times and mitigating congestion, and the underlying optimization is usually framed as a network design problem on top of a user-equilibrium traffic-assignment model [1], [2], [3]. In rapid transit and public transport systems, planners are often more concerned with coverage, accessibility, and social cohesion, and the relevant models combine line planning, frequency setting and infrastructure deployment under public-service criteria [4], [5], [6]. In contrast, privately operated transportation systems are mainly designed from a profitability viewpoint, where the network configuration must balance service attractiveness against operating and infrastructure costs, and where market capture against incumbent competitors plays a central role [7], [8]. From an optimization perspective, these settings all lead to network design problems,

but with markedly different objective functions and modeling choices.

Within this broad class of problems, hub-and-spoke networks play a particularly important role in air transportation [9], [10], [7], [8]. Structurally, they can be interpreted as the union of several star-shaped subnetworks interconnected through a reduced set of hub airports. Spoke airports mainly feed or receive local traffic, whereas hubs concentrate transfer activity and allow the operator to connect many origin-destination markets with a comparatively limited number of direct links. For airlines, this architecture is attractive because it consolidates traffic, improves the use of aircraft and airport resources, and allows broad market coverage without operating every origin-destination pair as a direct service. Consequently, hub-and-spoke network design for a private airline is fundamentally a profit-driven problem in which the key decisions concern hub activation and the assignment of capacities across airports and routes. Passenger flow variables are incorporated not as direct operational decisions, but as a modeling device to represent how users may react to the offered network and, therefore, how much revenue each market can generate for the operator.

Classical formulations address these decisions through mixed-integer programming (MIP) models [9], [10], [11], [12], [13], since fixed-cost activation decisions are naturally represented with binary variables. This is the standard modeling approach for hub location and network design, but the resulting formulations become combinatorially challenging as the number of airports, markets, and operating restrictions grows, with complexity increasing very rapidly once multicommodity interactions and fixed costs are included [11], [12]. Modeling passenger behavior creates an additional challenge. Simplified all-or-nothing assignment rules are linear and easy to embed in optimization models [1], [3], but they provide a poor representation of how users choose among competing itineraries. Logit-based demand models are considerably more realistic because they allow the captured share of each market to vary smoothly with the perceived utility of the available alternatives [14], [15]. Yet, when introduced into a network design problem, they typically give rise to nonlinear and nonconvex constraints. In practice these constraints are handled either through piecewise-linear mixed-integer approximations of the logit relation [16], which substantially inflate the number of binary variables and rapidly degrade the scalability of state-of-the-art branch-and-bound solvers, or through exponential-cone reformulations [17], which preserve convexity but remain numerically demanding because they couple every market

with cone constraints whose feasible set is hard to explore at scale. As a result, both approaches make medium- and large-size instances of the airline hub-and-spoke design problem computationally prohibitive in industrial settings.

This masters thesis builds on a recently proposed continuous optimization framework for transportation network design [6], [18], in which reweighted ℓ_1 sparsity is used to approximate the fixed costs associated with airport, hub, and route activation, while entropy-based regularization recovers logit market shares without explicitly imposing nonconvex share equations. On top of that continuous lower-level model we develop the main contribution of this work: we cast the airline design problem as a *bilevel* optimization model. The upper level represents the airline as a profit-maximizing decision-maker that selects market-specific coefficients, whereas the lower level returns the continuous network design induced by those coefficients. In this way, the calibration of the model is performed jointly with the network design, instead of being delegated to an exogenous heuristic. To solve the resulting bilevel problem we adopt a penalty-based reformulation in the spirit of recent first-order bilevel algorithms [19], [20], [21], combined with an outer iteration that updates the reweighted ℓ_1 terms, thereby improving the approximation of real fixed costs across successive solves.

The remainder of the paper is organized as follows. We first review the relevant literature on hub-and-spoke network design, continuous approximations of fixed-cost models, demand modeling, and bilevel optimization. We then describe the airline problem setting and present the mixed-integer baseline together with the continuous lower-level formulation. Next, we introduce the bilevel reformulation and the associated penalty-based solution strategy. Finally, we report computational experience on synthetic Spanish networks of increasing size and on a realistic Madrid-centred network with Iberia-style international connections, and close with the main conclusions.

II. STATE OF THE ART

The methodological contributions of this work touch four distinct bodies of literature. Because the paper proposes a continuous bilevel formulation of an airline hub-and-spoke design problem in which sparsity-inducing penalties replace integer activation variables and entropy regularization replaces the explicit logit constraint, the review is organized around those building blocks. We first cover, jointly, the classical hub-and-spoke design literature and the way it has historically represented airline demand. We then examine continuous and sparsity-based approximations, which underpin the lower-level model used in the rest of the paper. We close with a dedicated review of bilevel optimization and penalty methods, since the upper-level layer of the proposed model relies heavily on that machinery. The emphasis is placed throughout on hub-and-spoke airline applications, which is the application domain of this master's thesis.

A. Hub-and-Spoke Airline Network Design and Demand Modeling

Hub-and-spoke network design has traditionally been modeled as a discrete location-and-routing problem in which the main decisions are the selection of hub facilities, the assignment of spoke nodes to those hubs, and the activation of direct or inter-hub services. The foundational formulations cast these decisions as mixed-integer optimization problems in which the binary variables represent build/no-build choices and the continuous variables represent passenger or freight flows. The quadratic integer program proposed in [9] is generally regarded as the starting point of the modern hub-location literature, and the first integer linear formulation in which hub activation, spoke allocation and inter-hub flows are treated within a unified MIP model is the one of [10]. Subsequent works refined that template by exploring single-allocation versus multiple-allocation rules, by introducing tighter linearizations of the quadratic interaction terms [22], and by extending the framework to capacitated variants [23]. The state of the field after twenty-five years of activity is described in the comprehensive surveys [11], [12], [13], which catalog more than a hundred formulations differing in topology assumptions, capacity rules, cost structures, and solution techniques. A common thread in those reviews is that the underlying problems become very rapidly intractable as soon as fixed costs, multicommodity flows, route feasibility, and market-level economics are represented explicitly. In airline applications this combinatorial explosion is even more apparent, because the operator's profit ultimately depends on a large number of origin-destination markets that must be served simultaneously through shared hubs and shared route capacities. To enrich the airline-specific view, structural and competitive aspects of hub-and-spoke design have been studied in [7], and the operational and economic context of contemporary airline systems is summarized in [8], where the strategic role of hub airports for fleet utilization, market consolidation, and revenue generation is emphasized.

A second, methodologically intertwined, line of research concerns how passenger demand is represented inside these network design models. The simplest approach treats demand as fixed and routes it deterministically through the designed network. Such all-or-nothing assignment rules are linear, easy to embed into MIP formulations, and well suited for benchmark instances [1], [3]. They are, however, structurally unable to capture the way real passengers react to fares, travel times, and the quality of the offered service. Discrete-choice theory provides a much more behaviorally meaningful description of demand capture, and within that framework the multinomial logit model is by far the most widely used specification [14], [15]. Formally, each passenger in market (o, d) chooses an alternative $q \in \mathcal{Q}^{od}$ by maximizing a random utility

$$U_q^{od} = v_q^{od} + \varepsilon_q^{od}, \quad (1)$$

where the deterministic component v_q^{od} aggregates observable disutilities such as fare and time, $v_q^{od} = \omega_p p_q^{od} + \omega_t t_q^{od}$, and

the random component ε_q^{od} is assumed to be independent and identically distributed Gumbel. Under those assumptions, the probability that alternative q is selected in market (o, d) admits the closed-form expression

$$P^{od}(q) = \frac{\exp(v_q^{od})}{\sum_{q' \in Q^{od}} \exp(v_{q'}^{od})}, \quad (2)$$

which is the multinomial logit choice rule. When this expression is embedded in a network design problem so that the market shares depend on the actual network being designed, the resulting optimization model inherits nonlinear, nonconcave couplings between design variables, utilities, and demand allocation. For airline hub-and-spoke design this is precisely where realistic demand modeling starts to conflict with tractable large-scale optimization. Existing remedies typically replace the closed-form share (2) either by a piecewise-linear approximation supported on additional binary variables [16], or by an exponential-cone reformulation [17]. Both options are valid in principle, but both also inflate the model and quickly run into the same scalability barrier that affects the underlying mixed-integer hub design problem.

B. Continuous and Sparsity-Based Approximations

The computational difficulties listed above have motivated a different modeling philosophy in which some of the binary activation decisions are replaced by sparse continuous variables. Rather than deciding directly whether an airport, hub, or service is open, one optimizes continuous capacities and then promotes sparse solutions so that inactive components collapse naturally to zero. The conceptual starting point is the well-known reweighted ℓ_1 minimization framework, in which a nonconvex logarithmic surrogate of the ℓ_0 pseudonorm,

$$\kappa \log\left(1 + \frac{z}{\varepsilon}\right), \quad (3)$$

is iteratively linearized through a majorization-minimization argument, yielding at iteration k the convex weighted penalty $\kappa z / (z^{(k-1)} + \varepsilon)$ [24]. Small previous values of $z^{(k-1)}$ generate large weights and aggressively penalize new activation, whereas already-active components are penalized only mildly. Iterating this update progressively drives the unused infrastructure to zero, which makes the scheme a continuous proxy of the ℓ_0 activation cost. The same philosophy can be extended to the demand side: instead of imposing the logit expression (2) as an explicit nonconvex equation, the market split can be induced through smooth entropy regularization, which preserves convexity and recovers the logit shares as optimality conditions of the resulting smooth problem. This twofold strategy—sparsity for fixed costs and entropy for demand—has been recently advocated as a scalable alternative to mixed-integer formulations for transportation network design in [6], where it is applied to large rapid-transit network instances, and it has been extended to a single-level competitive airline hub-and-spoke design model in [18]. Although these ideas are well established in continuous optimization and signal

processing, they remain much less common in airline hub-and-spoke network design than the classical mixed-integer viewpoint.

C. Bilevel Optimization and Penalty Methods

The fourth ingredient of the proposed methodology is bilevel optimization, which is the natural mathematical framework whenever one layer of decisions is used to influence the solution of another. Following the standard convention, a bilevel program with upper-level decision u and lower-level decision y is written as

$$\max_{u \in \mathcal{U}} F(u, y) \quad (4)$$

$$\text{s.t. } y \in \arg \min_{y' \in Y(u)} f(u, y'), \quad (5)$$

where the upper-level player anticipates that the lower-level player will respond to any choice u with an optimal y . This hierarchical structure was first identified in the Stackelberg leader-follower models of competitive markets [25], [26], where one firm sets prices or quantities and the other responds optimally. The first formal optimization treatment of multilevel problems is generally attributed to [27], who showed how constraints can themselves be parametric optimization problems, and the operations research community has been studying their algorithmic and complexity properties ever since.

From the very beginning it was clear that bilevel programs are intrinsically hard. Even when both levels are linear, the feasible set of the upper-level player is in general nonconvex and disconnected, and the problem is strongly NP-hard. As a consequence, the literature has developed along several complementary directions. A first family of methods replaces the lower-level optimization by its Karush-Kuhn-Tucker (KKT) conditions, producing a single-level mathematical program with complementarity constraints whose binary indicators select the active inequalities. This route works particularly well when the lower-level problem is convex, and it has been widely used since the early references on quadratic bilevel programming. A second family adapts descent methods to the bilevel structure by approximating the hyperderivative through the implicit-function theorem; in recent years, these implicit-gradient descent (IGD) ideas have become the backbone of bilevel machine-learning algorithms [20], [21], [28]. A third family relies on evolutionary or metaheuristic strategies and is intended for problems whose mathematical structure is too unfavorable for derivative-based methods [29]. Standard survey references covering these developments include [30], [31], [32], [29].

Penalty-based reformulations occupy a central place in this landscape and are particularly relevant for the present work. The basic idea, due to Aiyoshi and Shimizu, is to replace the lower-level optimality condition by a penalty term added to the upper-level objective [33], [19]. When the lower level is unconstrained, a typical penalty is $\|\nabla_y g(u, y)\|^2$, so that large penalty weights drive the gradient of the lower-level objective to zero. When the lower level has constraints, the same idea is applied to the gradient of the associated Lagrangian

together with the remaining KKT conditions. An alternative formulation that is more convenient for the algorithms used in this work replaces the lower-level optimality by a *value-function* penalty, namely $g(u, y) - v(u)$, where $v(u)$ denotes the lower-level optimal value. Exact penalization results for this construction were established in [34], and recent first-order penalty algorithms have shown that constrained bilevel problems with convex lower-level structure can be approached through penalized single-level surrogates with implementable complexity guarantees [20], [21]. The algorithmic scheme developed in the present paper belongs to that family.

To close, it is worth stressing that the present paper lies at the intersection of all four strands above. Relative to the traditional hub-location literature, it avoids an explicit mixed-integer treatment of airport, hub, and service activation and instead works with a continuous sparse design model. Relative to standard deterministic network-design formulations, it incorporates endogenous demand capture through utility and entropy terms, so that market shares react to the designed network. Relative to generic sparsity-based continuous models, it is tailored to airline hub-and-spoke design and to a profit-oriented private operator. Finally, relative to the general bilevel literature, the upper level is not used here to redesign the network directly, but to calibrate the market-specific coefficients that guide the lower-level continuous approximation toward the airline's true profit objective. This combination of entropy-based demand modeling, sparse continuous activation, and value-function-penalty bilevel calibration is, to our knowledge, not standard in the airline hub-and-spoke design literature and defines the methodological contribution developed in the remainder of the paper.

III. PROBLEM SETTING

We consider the strategic design of a hub-and-spoke air transportation network for a private airline that enters a competitive market. The objective is to decide simultaneously which airports must be activated, which of those airports must additionally be upgraded to hubs, what spoke and hub capacities must be installed at each location, and what service capacity must be assigned to each candidate flight leg. Together, these decisions determine the topology of the network, its physical scale, and its operational footprint. Because the operator is a private firm rather than a public planner, all the design choices are driven by profit maximization: a network is attractive only insofar as the revenue extracted from passenger markets exceeds the cost of activating and operating the corresponding infrastructure.

Formally, let $\mathcal{S}$ denote the set of candidate airports, $\mathcal{A} \subseteq \mathcal{S} \times \mathcal{S}$ the set of candidate flight legs, and $\mathcal{K}$ the set of origin-destination markets with positive demand. Each market $(o, d) \in \mathcal{K}$ corresponds to a real travel need from origin o to destination d , characterized by a fixed demand volume and a fare charged by the airline. For every such market, the entrant competes against an outside alternative that aggregates incumbent airlines operating directly or with one stop, ground transportation modes when relevant, and the no-travel option.

The airline does not modify the offered alternatives of the outside option; it only decides whether and how to compete with them, both in terms of the network structure offered and in terms of the implicit pricing of time and fare embedded in passenger choice.

The network being designed follows a strict hub-and-spoke architecture. Some airports may be activated as pure spokes, in which case they only support local departures and arrivals to and from one or more hubs. Others may also receive additional transfer capacity and operate as hubs, processing connecting itineraries on top of their local activity. This distinction is central in the airline setting because local airport activity and transfer activity generate different physical resource requirements (gates, slots, ground handling, baggage transfer) and different unit costs. As a result, the design problem must decide not only which airports are active, but also which of them are upgraded to hubs, how much spoke capacity and how much hub capacity is installed at each airport, and how much service capacity is assigned to each candidate connection. The model also forces flight services to be symmetric, $A_{ij} = A_{ji}$, which is a stylized but realistic representation of the fact that aircraft must overnight at hubs and operate balanced rotations.

Passenger behavior is endogenous to the model. For every market (o, d) , the airline does not impose a captured volume; instead, it determines the demand share that the entrant captures and the itinerary used to serve that demand inside the designed network. Passenger flow variables are not interpreted as direct operational controls—the airline does not decide which specific passenger boards which specific flight—but as a modeling device through which the optimization model internalizes the interaction between network attractiveness and revenue generation. The captured share in a market depends on a perceived utility that combines the fare the airline charges and the total travel time of the chosen itinerary, and is compared with the analogous utility offered by the outside alternative. Whenever the airline cannot produce a sufficiently attractive utility, demand collapses to the outside alternative and the corresponding revenue is lost.

Beyond the airline-specific design choices, the model is subject to a number of structural restrictions that reflect the resource limits and the operational logic of a real airline. The total number of flights cannot exceed the available fleet, the local airport capacity must be sufficient to absorb the workload of departures and arrivals, the hub capacity at any airport must accommodate transfer passengers in addition to local traffic, and direct services between two airports are admissible only if at least one of the endpoints is operated as a hub. In addition, the model includes a budget restriction that bounds the total expenditure on airport activation, hub activation and installed capacities. Together with the fleet limit and the capacity restrictions, this budget constraint represents the strategic envelope inside which the airline must build a profitable competitive network. The remainder of this section formalizes these elements as a mixed-integer optimization model that will serve as a baseline for the continuous bilevel reformulation developed in the rest of the paper.

A. Sets, Parameters, and Variables

To state the model precisely, we use the following notation.

Sets:

- $\mathcal{S}$ is the set of candidate airports.
- $\mathcal{A} \subseteq \mathcal{S} \times \mathcal{S}$ is the set of candidate flight legs.
- $\mathcal{K}$ is the set of origin-destination markets with positive demand.
- $\mathcal{Q}_{\text{ext}}^{od}$ is the set of outside alternatives available in market (o, d) .
- $\mathcal{N}_{\text{in}}(i)$ and $\mathcal{N}_{\text{out}}(i)$ denote the incoming and outgoing neighbors of airport i .

Parameters:

- w^{od} is the demand in market (o, d) and p^{od} is the fare charged by the entrant in that market.
- v_q^{od} is the utility of outside alternative $q \in \mathcal{Q}_{\text{ext}}^{od}$, with $p_{\text{ext},q}^{od}$ and $t_{\text{ext},q}^{od}$ denoting the corresponding fare and travel time.
- t_{ij} is the travel time associated with leg (i, j) , and $t_{ij,\text{trans}}^{od}$ is the transfer time induced at the connecting airport when leg (i, j) is used inside the itinerary of market (o, d) .
- c_{s_i} and c_{h_i} are the fixed costs of activating airport i as a spoke and as a hub, respectively.
- $\bar{c}_{s_i}$ is the variable cost of installing airport capacity at node i , $\bar{c}_{a_{ij}}$ is the variable cost of installing service capacity on leg (i, j) , and $c_{o_{ij}}$ is the operating cost of leg (i, j) .
- a_{nom} is the nominal aircraft capacity, τ is the route load factor, and σ and σ_i are scaling coefficients for transfer and airport processing capacity.
- λ weights hub fixed costs inside the budget, a_{max} is the total fleet limit, b is the available budget, and M is a sufficiently large constant.
- ω_p and ω_t are the coefficients associated with fare and travel time in the passenger utility model, and $n_{\text{airlines}} = |\mathcal{Q}_{\text{ext}}^{od}|$ is the number of outside alternatives used in the entropy regularizer.
- ε_S and ε_A are small positive smoothing constants used in the reweighted ℓ_1 approximation of station and arc fixed costs, respectively.

Decision variables:

- $S_i^b \in \{0, 1\}$ indicates whether airport i is active, and $S_i^{hb} \in \{0, 1\}$ indicates whether airport i operates as a hub.
- $S_i \geq 0$ and $S_i^h \geq 0$ are the spoke and hub capacities installed at airport i .
- $A_{ij} \geq 0$ is the service capacity assigned to leg (i, j) .
- $F^{od} \in [0, 1]$ is the share of market (o, d) captured by the entrant.
- F_{ij}^{od} indicates whether the itinerary used by market (o, d) traverses leg (i, j) .
- Z^{od} indicates whether market (o, d) is served by the entrant.

For compactness, the design variables are grouped as

$$\mathcal{X}_{\text{ser}} = \{S_i^b, S_i^{hb}, S_i, S_i^h, A_{ij}\}, \quad (6)$$

whereas the passenger and routing variables are grouped as

$$\mathcal{X}_{\text{pax}} = \{F^{od}, F_{ij}^{od}, Z^{od}\}. \quad (7)$$

B. Mixed-Integer Viewpoint

To motivate the continuous reformulation developed later, we first describe the underlying mixed-integer viewpoint. In this section, capital letters denote the classical design variables: S_i and S_i^h are the spoke and hub capacities installed at airport i , A_{ij} is the service capacity assigned to leg (i, j) , S_i^b indicates whether airport i is active, and S_i^{hb} indicates whether airport i operates as a hub. For passenger assignment, F^{od} denotes the share of market (o, d) captured by the entrant, and F_{ij}^{od} indicates whether the itinerary used by market (o, d) traverses leg (i, j) .

Since the operator is private, the natural objective is profit maximization. Equivalently, the model can be written as the minimization of negative profit:

$$\min - \sum_{(o,d) \in \mathcal{K}} w^{od} p^{od} F^{od} + \sum_{(i,j) \in \mathcal{A}} c_{o_{ij}} A_{ij}, \quad (8)$$

where the first term represents passenger revenue and the second term represents the operating cost of the flights offered by the entrant. The design therefore balances market capture against the cost of supplying the corresponding air service.

C. Constraint Families

The mixed-integer viewpoint combines infrastructure activation decisions, service-capacity constraints, passenger-routing constraints, and demand-capture constraints. For clarity, we discuss these restrictions by family.

1) *Airport Activation and Budget Constraints:* The hub-and-spoke structure requires two airport-capacity variables. Spoke capacity S_i accounts for local departures and arrivals, whereas hub capacity S_i^h accounts for transfer activity. Their binary activation variables satisfy

$$S_i \leq M S_i^b \quad \forall i \in \mathcal{S}, \quad (9)$$

$$S_i^h \leq M S_i^{hb} \quad \forall i \in \mathcal{S}, \quad (10)$$

$$S_i^{hb} \leq S_i^b \quad \forall i \in \mathcal{S}, \quad (11)$$

Constraint (9) activates spoke capacity only when airport i is opened, and (10) plays the same role for hub capacity. Constraint (11) enforces the hierarchical logic of the design: an airport can operate as a hub only if it is already active in the network. In contrast, we do not impose a big- M activation relation on route capacities, because the current airline formulation does not associate fixed costs with arcs.

The infrastructure budget must account for spoke activation, hub activation, and installed airport capacity. This leads to

$$\sum_{i \in \mathcal{S}} c_{s_i} S_i^b + \lambda \sum_{i \in \mathcal{S}} c_{h_i} S_i^{hb} + \sum_{i \in \mathcal{S}} \bar{c}_{s_i} (S_i + S_i^h) \leq b, \quad (12)$$

where λ modulates the relative cost of opening hubs. This restriction captures the strategic trade-off between enlarging the airport system and preserving enough resources to support flight operations. The total number of flights is also limited by the available fleet:

$$\sum_{(i,j) \in \mathcal{A}} A_{ij} \leq a_{\max}. \quad (13)$$

Constraint (13) prevents the model from assigning more service capacity than can be supported by the available aircraft. Because aircraft are assumed to overnight at hubs, the service pattern must also be symmetric, which yields

$$A_{ij} = A_{ji} \quad \forall (i,j) \in \mathcal{A}. \quad (14)$$

This symmetry condition is a stylized representation of the hub-and-spoke operating pattern and avoids one-way capacity imbalances that would be unrealistic in the strategic design stage.

2) *Leg and Airport Capacity Constraints:* Service capacity, airport capacity, and transfer capacity are jointly constrained by the hub-and-spoke logic. The capacity of each leg must be sufficient to transport the demand assigned to it:

$$\sum_{(o,d) \in \mathcal{K}} w^{od} F^{od} F_{ij}^{od} \leq a_{\text{nom}} \tau A_{ij} \quad \forall (i,j) \in \mathcal{A}, \quad (15)$$

where a_{nom} is the nominal aircraft capacity and τ is the load factor. Constraint (15) therefore ensures that the aggregate number of passengers routed through leg (i,j) does not exceed the effective transport capacity supplied on that leg.

Local airport capacity is enforced through the aggregate arrival and departure workload,

$$\sigma_i \sum_{j \in \mathcal{N}_{\text{in}}(i)} A_{ji} + \sigma_i \sum_{j \in \mathcal{N}_{\text{out}}(i)} A_{ij} \leq S_i + S_i^h \quad \forall i \in \mathcal{S}, \quad (16)$$

which states that the total local workload induced by incoming and outgoing flights must be supported by the installed airport capacity. Since hubs process not only local traffic but also connecting passengers, transfer activity is limited separately through

$$\sigma \left(\sum_{j \in \mathcal{N}_{\text{out}}(i)} \sum_{\substack{(o,d) \in \mathcal{K} \\ o \neq i}} \frac{w^{od} F^{od} F_{ij}^{od}}{a_{\text{nom}} \tau} + \sum_{j \in \mathcal{N}_{\text{in}}(i)} \sum_{\substack{(o,d) \in \mathcal{K} \\ d \neq i}} \frac{w^{od} F^{od} F_{ji}^{od}}{a_{\text{nom}} \tau} \right) \leq S_i^h \quad \forall i \in \mathcal{S}. \quad (17)$$

Constraint (17) captures the fact that connecting passengers can only be processed if the corresponding airport has enough hub capacity. This is one of the defining structural features of the model, because it distinguishes the role of a spoke from the role of a hub. In addition, a direct service between the endpoints of a market is allowed only if at least one of those endpoints is operated as a hub:

$$\sigma \frac{w^{od} F^{od} F_{od}^{od}}{a_{\text{nom}} \tau} \leq S_o^h + S_d^h \quad \forall (o,d) \in \mathcal{K}. \quad (18)$$

Constraint (18) prevents the model from generating direct services that are incompatible with the hub-and-spoke architecture.

3) *Passenger Routing and Market Feasibility Constraints:* The passenger flow variables must satisfy the usual conservation equations. If Z^{od} denotes whether market (o,d) is served by the entrant, then

$$\sum_{j \in \mathcal{N}_{\text{out}}(i)} F_{ij}^{od} - \sum_{j \in \mathcal{N}_{\text{in}}(i)} F_{ji}^{od} = \begin{cases} Z^{od}, & i = o, \\ -Z^{od}, & i = d, \\ 0, & \text{otherwise,} \end{cases} \quad (19)$$

for every $i \in \mathcal{S}$ and $(o,d) \in \mathcal{K}$. Constraint (19) ensures that each served market has a valid path from its origin to its destination and that no artificial creation or destruction of flow occurs at intermediate nodes.

To avoid attracting markets that are not profitable, the revenue collected on each served market must dominate the operating cost induced by its itinerary:

$$w^{od} p^{od} F^{od} \geq \sum_{(i,j) \in \mathcal{A}} \frac{w^{od} c_{oij} F^{od} F_{ij}^{od}}{a_{\text{nom}} \tau} \quad \forall (o,d) \in \mathcal{K}. \quad (20)$$

Constraint (20) filters out unattractive itineraries by requiring each captured market to be self-supporting from the airline's viewpoint.

4) *Demand-Capture Constraint:* Passenger choice is represented through a logit-type capture rule. The utility of the entrant in market (o,d) combines the fare charged by the airline and the total travel time of the selected itinerary, namely

$$u_{\text{self}}^{od} = \omega_p p^{od} + \omega_t \sum_{(i,j) \in \mathcal{A}} t_{ij} F_{ij}^{od}. \quad (21)$$

The captured share is then constrained by

$$F^{od} \leq \frac{\exp(u_{\text{self}}^{od})}{\exp(u_{\text{self}}^{od}) + \sum_{q \in \mathcal{Q}_{\text{ext}}^{od}} \exp(u_q^{od})} \quad \forall (o,d) \in \mathcal{K}, \quad (22)$$

Constraint (22) couples network design with passenger behavior in a nonconvex way. The share captured by the entrant depends on the utility generated by the network that has just been designed, while that same utility depends on the routing pattern selected for the market.

D. Compact MIP Statement and Computational Limitations

Putting all the constraint families together, the airline hub-and-spoke network design problem can be written compactly as

$$\begin{aligned} \min_{\mathcal{X}_{\text{ser}}, \mathcal{X}_{\text{pax}}} \quad & - \sum_{(o,d) \in \mathcal{K}} w^{od} p^{od} F^{od} + \sum_{(i,j) \in \mathcal{A}} c_{oij} A_{ij} \\ \text{s.t.} \quad & (9)-(11), (12)-(14), \\ & (15)-(18), (19)-(20), (21)-(22), \\ & S_i^b, S_i^{hb}, Z^{od} \in \{0,1\}, S_i, S_i^h, A_{ij} \geq 0, F^{od} \in [0,1]. \end{aligned} \quad (23)$$

Problem (23) is a mixed-integer nonlinear program with three sources of structural difficulty. First, the activation variables

S_i^b , S_i^{hb} , and the per-market service indicator Z^{od} generate a binary count that grows linearly with the number of candidate airports and quadratically with the number of markets. Second, the routing constraints couple capacity and flow variables through the bilinear products $F^{od}F_{ij}^{od}$, which already make the feasible set nonconvex even before the demand-capture constraint enters the picture. Third, the logit equation (22) introduces the exponential structure of multinomial choice into an otherwise mixed-integer model, breaking convexity at the very point where the model becomes economically meaningful.

In practice, the standard remedy is to replace the closed-form logit relation by a piecewise-linear approximation of either the right-hand side of (22) or the underlying log-sum-exp utility aggregator [16], which yields an MILP that can in principle be solved by a state-of-the-art branch-and-bound code. Each piece of the linearization, however, requires its own auxiliary binary variable per market, and the number of pieces needed to obtain an accurate approximation grows quickly with the demand and utility ranges of the instance. As a consequence, the binary count of the MILP scales not only with the network size but also with the desired fidelity of the demand model, and the resulting solver branch tree becomes computationally prohibitive far before instances of practical industrial size are reached. Convex exponential-cone reformulations [17] avoid the binary inflation but couple every market through cone constraints whose interior-point exploration is itself expensive, so they are likewise unable to scale to the airline instances that motivate this work. Empirically, even moderate Spanish synthetic networks of six to eight airports already push commercial branch-and-bound solvers into the multi-hour regime, while real Iberia-style international instances with several dozen candidate airports lie well beyond what a direct mixed-integer solve can handle in any reasonable time. The next section addresses these computational obstacles by replacing the explicit nonlinear demand-capture equation by an equivalent continuous viewpoint, in which fixed costs are approximated through sparsity-inducing penalties and the logit split is recovered through entropy regularization.

IV. CONTINUOUS AND BILEVEL FORMULATION

This section presents the central methodological contribution of the paper. Starting from the mixed-integer nonlinear model introduced in the previous section, we build a continuous approximation that remains faithful to the economics of the airline design problem but whose computational cost is dominated by convex subproblems rather than by an exponentially growing branch-and-bound tree. The resulting lower-level model is convex for fixed input parameters and is therefore amenable to off-the-shelf interior-point solvers [6], [18]. On top of that continuous approximation we then introduce a second decision layer whose role is to *calibrate*, rather than re-design, the way the lower-level model values each market. This second layer is what turns the model into a bilevel program.

The motivation for that second layer comes from an inherent limitation of the continuous lower-level formulation. The continuous model captures network design, capacity allocation and behaviorally meaningful market shares with a single convex problem, but it requires the modeller to choose how much importance each origin-destination market should have inside the lower-level objective. Different choices of these market-specific weights lead to very different network designs and, more importantly, to very different airline profits, even when the underlying constraints and resources are kept fixed. In a small benchmark instance the weights can be tuned by hand, but in a realistic airline planning context, where the number of markets is in the dozens or hundreds, that manual calibration becomes infeasible. The natural alternative is to make these weights a decision variable of the model itself, and to choose them so that the resulting design maximizes the airline's true profit. This is precisely the bilevel structure that we adopt: the lower level returns the continuous design induced by a given set of market weights, and the upper level selects those weights so that the airline-level profit is as large as possible.

Viewed in simple terms, the proposed method combines two nested tasks that interact through the market-weight variables. The first task takes a set of market weights as input and produces, as output, a hub-and-spoke network, its installed capacities, the routing pattern of the captured demand, and the market shares induced by the resulting service offer under a smooth behavioral model. This task plays the role of an experienced planning department that, given the strategic priorities of the airline, returns a coherent operational design. The second task observes the result of the first one and modifies the strategic priorities, that is, the market weights themselves, in order to push the design toward higher airline profit. This task plays the role of a senior decision-maker who is not interested in flight schedules per se, but who decides how much each individual market should weigh in the planning department's objective. The fact that the second task does not directly choose the network but only the criteria under which the network is chosen is what makes the model bilevel.

The remainder of the section is organized as follows. We first present the continuous lower-level model that is solved inside the nested scheme, redefine the variables and parameters used by the bilevel layer, and explain how entropy regularization recovers a logit-style market split through a convex-conjugate construction. We then introduce the reweighted ℓ_1 approximation of fixed activation costs that is used to keep deployment decisions sparse. With those two ingredients in place we state the complete bilevel problem, including its objective and its constraint set. The penalty-based reformulation of the bilevel problem and the associated solution algorithm are deferred to the next section.

A. Continuous Design Model

Before stating the continuous lower-level model, we make explicit the change of perspective with respect to the mixed-integer formulation of Section III. The continuous model

retains the same set of physical decisions, namely airport capacities, hub capacities, link capacities, market shares and routing variables, but represents them through nonnegative continuous variables only. Integer activation variables are removed, and the build/no-build logic that they implement is recovered through a sparsity-inducing penalty on the capacity variables themselves. The same is done with the route-use indicators F_{ij}^{od} , which are relaxed from $\{0,1\}$ to $[0,1]$ and reinterpreted as the fraction of market (o,d) that is routed through leg (i,j) . Finally, the binary service indicator Z^{od} disappears altogether, because the flow-conservation equations are written directly in terms of the captured share F^{od} .

Concretely, the continuous design variables are grouped in the vector

$$X := (S, S^h, A, F, F^{\text{ext}}, \{F_{ij}^{od}\}_{(i,j),(o,d)}), \quad (24)$$

where $S_i, S_i^h \geq 0$ are the spoke and hub capacities at airport i , $A_{ij} \geq 0$ is the service capacity assigned to leg (i,j) , $F^{od} \in [0,1]$ and $F^{\text{ext},od} \in [0,1]$ are the market shares of the entrant and the outside alternative for market (o,d) , and $F_{ij}^{od} \in [0,1]$ are the route-share variables. All other parameters of the model—demand w^{od} , fare p^{od} , travel time t_{ij} , operating cost c_{oij} , fixed costs c_{si}, c_{hi} , variable capacity cost $\bar{c}_{si}$, fleet limit $a_{\max}$, budget b , scaling coefficients $\sigma, \sigma_i, a_{\text{nom}}, \tau$, utility coefficients ω_p, ω_t , hub-weight λ , smoothing constants $\varepsilon_S, \varepsilon_A$, and outside utilities v_q^{od} —coincide with those introduced in Section III and keep their original meaning.

In addition to those quantities, the bilevel layer introduces a pair of market-specific scalar coefficients $\hat{\alpha}^{od} > 0$ and $\hat{\beta}^{od} > 0$ for every market $(o,d) \in \mathcal{K}$. These coefficients do not modify the feasible set of the lower-level problem; their only role is to modulate the contribution of each market to the lower-level objective. Following the parametric form used in the implementation [18], the lower-level *market weight* for market (o,d) is defined as

$$\varphi^{od}(\hat{\alpha}, \hat{\beta}) := \hat{\alpha}^{od} (p^{od} w^{od})^{\hat{\beta}^{od}}, \quad (25)$$

so that the scaling factor $\hat{\alpha}^{od}$ controls the linear sensitivity to the market and $\hat{\beta}^{od}$ controls how strongly large-revenue markets are emphasized relative to small-revenue ones. Both coefficients are allowed to vary within bounded intervals, which will be made explicit below when the bilevel problem is stated.

For fixed market coefficients, the continuous design model is

$$v(\hat{\alpha}, \hat{\beta}) := \min_{X \in \mathcal{X}_c} \Phi_c(X; \hat{\alpha}, \hat{\beta}), \quad (26)$$

where $\mathcal{X}_c$ denotes the continuous feasible set and the lower-level objective is

$$\begin{aligned} \Phi_c(X; \hat{\alpha}, \hat{\beta}) := & \text{OP}(A) + \text{DEP}(S, S^h, A) \\ & + \text{U}_{\text{self}}(X; \hat{\alpha}, \hat{\beta}) + \text{U}_{\text{ext}}(X; \hat{\alpha}, \hat{\beta}) \\ & + \text{H}_{\text{self}}(X; \hat{\alpha}, \hat{\beta}) + \text{H}_{\text{ext}}(X; \hat{\alpha}, \hat{\beta}). \end{aligned} \quad (27)$$

The six terms have, respectively, the meaning of an operating cost, a sparse deployment cost that approximates the fixed-cost budget, a utility term for the entrant, a utility term for the outside alternative, and two entropy terms that regularize the market split. The deployment term is introduced in Section IV-B; the remaining four terms are detailed below.

a) *Operating costs.*: The direct flight operating cost is

$$\text{OP}(A) = \sum_{(i,j) \in \mathcal{A}} c_{oij} A_{ij}. \quad (28)$$

This term penalizes dense or high-capacity route structures and preserves the economic trade-off between market attractiveness and network operating effort.

b) *Utility terms.*: For market (o,d) , the systematic utility of the entrant is represented by

$$u_{\text{self}}^{od}(X) := \omega_p p^{od} + \omega_t \sum_{(i,j) \in \mathcal{A}} t_{ij} F_{ij}^{od}, \quad (29)$$

where ω_p and ω_t measure the sensitivity to fare and travel time. The utility contribution of the entrant in the lower-level objective is then written as

$$\text{U}_{\text{self}}(X; \hat{\alpha}, \hat{\beta}) = - \sum_{(o,d) \in \mathcal{K}} \varphi^{od} u_{\text{self}}^{od}(X) F^{od}, \quad (30)$$

while the outside alternative contributes

$$\text{U}_{\text{ext}}(X; \hat{\alpha}, \hat{\beta}) = - \sum_{(o,d) \in \mathcal{K}} \varphi^{od} v_{\text{ext}}^{od} F^{\text{ext},od}. \quad (31)$$

Since the lower level is a minimization problem, the negative sign in both terms means that allocations with larger utility are preferred. The market weight φ^{od} determines how strongly that preference influences the resulting network.

c) *Entropy regularization and endogenous logit shares.*: The continuous formulation also includes

$$\text{H}_{\text{self}}(X; \hat{\alpha}, \hat{\beta}) = \sum_{(o,d) \in \mathcal{K}} \varphi^{od} F^{od} (\log(F^{od}) - 1), \quad (32)$$

$$\text{H}_{\text{ext}}(X; \hat{\alpha}, \hat{\beta}) = \sum_{(o,d) \in \mathcal{K}} \varphi^{od} F^{\text{ext},od} \left(\log \left(\frac{F^{\text{ext},od}}{n_{\text{airlines}}} \right) - 1 \right). \quad (33)$$

These terms are much more than a smoothing device. They are precisely what replaces the explicit nonlinear logit equation (22) of the mixed-integer model by a convex variational representation of passenger choice. To make this point explicit, consider the simplex

$$\Delta^{od} := \{(F^{od}, F^{\text{ext},od}) \in \mathbb{R}_+^2 : F^{od} + F^{\text{ext},od} = 1\}, \quad (34)$$

which represents the set of admissible market splits in market (o,d) . Inside Δ^{od} , the utility terms (30) and (31) favor the alternative with the highest perceived utility, whereas the entropy terms (32) and (33) penalize overly concentrated allocations and create a strictly convex trade-off. The combined effect is a smooth market split that avoids unrealistic all-or-nothing assignments and recovers the same qualitative logic as the logit model (2).

The formal justification of this construction is the convex-conjugate relation between the log-sum-exp aggregator and negative entropy [6]. For the choice probability vector $F \in \Delta^{od}$ and the deterministic utility vector v , the log-sum-exp function and its convex conjugate (the negative entropy) form a Legendre dual pair,

$$f(v) = \log \sum_q \exp(v_q), \quad f^*(F) = \sum_q F_q \log F_q, \quad (35)$$

with the convention that $0 \log 0 = 0$. A central property of convex conjugates is that, on the relative interior of their domains, the gradients ∇f and ∇f^* are inverse maps. In our setting, ∇f evaluated at the utility vector v returns exactly the logit choice probabilities (2), while ∇f^* evaluated at the share vector F returns the vector of logarithmic shares $\log F$. By Fenchel duality, the maximization

$$\max_{F \in \Delta^{od}} \sum_q F_q v_q^{od} - \sum_q F_q \log F_q \quad (36)$$

admits the log-sum-exp function as its optimal value and the logit probabilities as its unique maximizer. Changing signs to fit the minimization structure of the lower level, the maximization (36) becomes precisely the combination of utility and entropy terms written above. Therefore, the market shares are not imposed externally; they arise endogenously as the unique solution of the lower-level problem in the absence of binding capacity constraints. This convex-conjugate viewpoint is the same one that underpins the equivalence theorem in [6]: when the capacity constraints of the continuous design problem are inactive at the optimum, the entropy-regularized passenger-flow solution is an optimal solution of the corresponding mixed-integer passenger-assignment problem. The continuous model therefore preserves a behaviorally meaningful demand split while remaining smooth, convex, and computationally tractable.

d) Structural constraints.: The continuous feasible set $\mathcal{X}_c$ is defined by the structural restrictions of the hub-and-spoke problem, with the binary decisions replaced by sparse capacities and the routing variables relaxed to continuous values. Specifically, $\mathcal{X}_c$ collects the link-capacity, station-

capacity, hub-capacity, and direct-hub restrictions

$$\sum_{(o,d) \in \mathcal{K}} w^{od} F_{ij}^{od} \leq a_{\text{nom}} \tau A_{ij} \quad \forall (i,j) \in \mathcal{A}, \quad (37)$$

$$\sigma \left(\sum_j A_{ji} + \sum_j A_{ij} \right) \leq S_i + S_i^h \quad \forall i \in \mathcal{S}, \quad (38)$$

$$\sigma \left(\sum_{\substack{(o,d),j \\ o \neq i}} \frac{w^{od} F_{ij}^{od}}{a_{\text{nom}} \tau} + \sum_{\substack{(o,d),j \\ d \neq i}} \frac{w^{od} F_{ji}^{od}}{a_{\text{nom}} \tau} \right) \leq S_i^h \quad \forall i \in \mathcal{S}, \quad (39)$$

$$\sigma \frac{w^{od} F_{od}^{od}}{a_{\text{nom}} \tau} \leq S_o^h + S_d^h \quad \forall (o,d) \in \mathcal{K}, \quad (40)$$

$$A_{ij} = A_{ji} \quad \forall (i,j) \in \mathcal{A}, \quad (41)$$

$$\sum_{(i,j) \in \mathcal{A}} A_{ij} \leq a_{\text{max}}, \quad (42)$$

together with flow-conservation and market-split restrictions,

$$\sum_j F_{oj}^{od} - \sum_i F_{io}^{od} = F^{od} \quad \forall (o,d) \in \mathcal{K}, \quad (43)$$

$$\sum_j F_{dj}^{od} - \sum_i F_{id}^{od} = -F^{od} \quad \forall (o,d) \in \mathcal{K}, \quad (44)$$

$$\sum_j F_{ij}^{od} - \sum_j F_{ji}^{od} = 0 \quad \forall i \notin \{o,d\}, (o,d) \in \mathcal{K}, \quad (45)$$

$$F^{od} + F^{\text{ext},od} = 1 \quad \forall (o,d) \in \mathcal{K}, \quad (46)$$

$$w^{od} p^{od} F^{od} \geq \sum_{(i,j) \in \mathcal{A}} \frac{w^{od} c_{oij} F_{ij}^{od}}{a_{\text{nom}} \tau} \quad \forall (o,d) \in \mathcal{K}, \quad (47)$$

and the nonnegativity and box restrictions

$$\begin{aligned} 0 &\leq F_{ij}^{od} \leq 1 && \forall (i,j) \in \mathcal{A}, (o,d) \in \mathcal{K}, \\ 0 &\leq F^{od} \leq 1, && 0 \leq F^{\text{ext},od} \leq 1 && \forall (o,d) \in \mathcal{K}, \\ &S_i, S_i^h &\geq 0 && \forall i \in \mathcal{S}, \\ &A_{ij} &\geq 0 && \forall (i,j) \in \mathcal{A}. \end{aligned} \quad (48)$$

Comparing with the mixed-integer formulation (23), three structural differences are apparent. First, the binary activation variables S_i^b, S_i^{hb}, Z^{od} have disappeared, together with their big- M activation constraints; their build/no-build effect is taken over by the deployment term DEP inside the objective and by the natural behavior of sparsity-inducing penalties. Second, the route-use variables are continuous and convexly

relaxed, which makes the flow-conservation equations linear and removes the bilinear terms $F^{od}F_{ij}^{od}$ that affected the mixed-integer leg-capacity constraints. Third, the explicit nonconvex logit equation (22) is no longer needed, because the combination of utility and entropy terms in the objective produces the same multinomial-logit shares through optimality conditions. The resulting model is convex for fixed market coefficients $(\hat{\alpha}, \hat{\beta})$, and the only remaining piece needed to specify it completely is the sparse deployment term DEP.

B. Reweighted ℓ_1 Approximation of Fixed Costs

The original airline design problem contains fixed costs associated with activating airports and hubs. Those costs are naturally related to an ℓ_0 -type activation mechanism: once a facility is used, a discrete cost must be paid. The ℓ_0 counting pseudonorm,

$$\|z\|_0 := \sum_i \mathbf{1}_{\{z_i \neq 0\}}, \quad (49)$$

is the exact mathematical object that captures this build/no-build accounting, but it is discontinuous and nonconvex, and it is not amenable to gradient-based optimization. The best plain convex surrogate of $\|\cdot\|_0$ is the ℓ_1 norm $\|z\|_1 = \sum_i |z_i|$, but on its own the ℓ_1 penalty does not adapt to the magnitude of the underlying decision: it charges every unit of capacity at the same rate, and it cannot reproduce the qualitatively different behavior of fixed versus variable cost contributions.

To circumvent these limitations, we approximate fixed costs through a reweighted ℓ_1 procedure derived from a logarithmic surrogate of $\|\cdot\|_0$ [24]. For a generic nonnegative scalar capacity $z \geq 0$, we consider

$$g(z) = \kappa \log\left(1 + \frac{z}{\varepsilon}\right), \quad \varepsilon > 0, \kappa > 0. \quad (50)$$

The function (50) is concave on $\mathbb{R}_+$, strictly increasing, and behaves very differently from a linear penalty in the two regimes of interest. Near the origin, $g(z) \approx (\kappa/\varepsilon)z$, so that small new capacities are charged a steep marginal price, mimicking the fact that turning on a previously inactive variable should be expensive. Far from the origin, $g(z) \approx \kappa \log(z/\varepsilon)$, so that already-active capacities are charged a much milder marginal price, which is consistent with the intuition that scaling up an already-active resource should not cost as much as turning it on. The function therefore behaves as a smooth interpolation between an ℓ_0 activation cost and a linear capacity cost.

Because g is concave, it admits a majorization at any iterate $z^{(k-1)} \geq 0$, namely

$$g(z) \leq g(z^{(k-1)}) + g'(z^{(k-1)})(z - z^{(k-1)}), \quad (51)$$

which, dropping constants, yields the weighted linear term

$$\tilde{g}_k(z) = \kappa \frac{z}{z^{(k-1)} + \varepsilon}. \quad (52)$$

Replacing the nonconvex log surrogate (50) by its majorant (52) and minimizing iteratively defines a majorization-minimization (MM) algorithm [24], [6]. Each MM iteration is a convex problem in which small previous values $z^{(k-1)} \approx 0$

produce large weights and encourage sparsity, while large previous values produce moderate weights and let active components grow.

Once the MM iteration has converged ($z \approx z^{(k-1)}$), two regimes emerge naturally:

- active components, with $z \gg \varepsilon$, are charged $\tilde{g}_k(z) \approx \kappa z/z = \kappa$, i.e. the fixed activation cost;
- pruned components, with $z = 0$, are charged $\tilde{g}_k(0) = 0$ regardless of the reference value.

In other words, the reweighted ℓ_1 regularizer behaves as a convex continuous proxy of the ℓ_0 activation pseudonorm: fixed costs are paid if the capacity is active and avoided if it is pruned, while ordinary variable capacity costs remain linear and additive on top.

In the proposed airline model this mechanism is applied to airport and hub capacities, which dominate the fixed-cost nonsmoothness of the formulation. Concretely, the deployment term in the lower-level objective (27) reads

$$\begin{aligned} \text{DEP}(S, S^h, A) = & \sum_{i \in S} \frac{c_{s_i}(S_i + S_i^h)}{\bar{S}_i + \bar{S}_i^h + \varepsilon_S} + \sum_{i \in S} \lambda \frac{c_{h_i}S_i^h}{\bar{S}_i^h + \varepsilon_S} \\ & + \sum_{i \in S} \bar{c}_{s_i}(S_i + S_i^h) + \sum_{(i,j) \in A} \bar{c}_{a_{ij}}A_{ij}, \end{aligned} \quad (53)$$

where $\bar{S}_i, \bar{S}_i^h, \bar{A}_{ij} \geq 0$ are the reference capacities maintained by the outer MM loop and $\varepsilon_S, \varepsilon_A > 0$ are the smoothing constants of the surrogate. The first two summands implement the reweighted ℓ_1 activation cost for airports and hubs, the third summand is the linear variable cost of installed airport capacity, and the fourth summand is the linear variable cost of installed arc capacity. As the outer iteration proceeds, the reference capacities $\bar{S}_i, \bar{S}_i^h$ are refreshed from the latest lower-level solution and the deployment term becomes a progressively tighter approximation of the underlying ℓ_0 fixed-cost structure. With the deployment term (53) the continuous lower-level model (26)–(48) is fully specified, and the budget restriction of the original mixed-integer problem is replaced by the penalty DEP inside the objective. This is the only remaining piece needed to state the bilevel problem in its complete form.

C. Complete Bilevel Problem

With the continuous lower-level model fully specified, the bilevel layer can now be defined. The role of the upper level is to choose the market coefficients $(\hat{\alpha}^{od}, \hat{\beta}^{od})_{(o,d) \in \mathcal{K}}$ so that the design returned by the lower level performs as well as possible under the true airline profit function

$$\Pi(X) := \sum_{(o,d) \in \mathcal{K}} p^{od} w^{od} F^{od} - \sum_{(i,j) \in A} c_{o_{ij}} A_{ij}. \quad (54)$$

The expression (54) is independent of the market weights φ^{od} , so that the upper-level objective measures profit on the same economic terms used by the airline itself, regardless of how

the lower-level model values each market internally. The set of admissible market coefficients is restricted to the box

$$\mathcal{H} := \{(\hat{\alpha}, \hat{\beta}) : \underline{\alpha} \leq \hat{\alpha}^{od} \leq \bar{\alpha}, \underline{\beta} \leq \hat{\beta}^{od} \leq \bar{\beta}, \forall (o, d) \in \mathcal{K}\}, \quad (55)$$

whose lower and upper bounds are chosen to keep φ^{od} in an economically meaningful range and to prevent the iterative updates from drifting toward degenerate values.

Putting the upper and lower levels together, the complete bilevel problem reads

$$\max_{(\hat{\alpha}, \hat{\beta}) \in \mathcal{H}, X} \Pi(X) \quad (56)$$

$$\text{s.t. } X \in \arg \min_{Y \in \mathcal{X}_c} \Phi_c(Y; \hat{\alpha}, \hat{\beta}), \quad (57)$$

with $\mathcal{X}_c$ collecting the structural constraints (37)–(48), the lower-level objective Φ_c given by (27) with the deployment term (53), and the bilevel feasibility expressed implicitly through the lower-level optimality condition (57). Problem (56)–(57) therefore couples three families of decisions: (i) the strategic market priorities $(\hat{\alpha}, \hat{\beta})$, chosen by the upper level inside the box $\mathcal{H}$; (ii) the physical design variables S, S^h, A , returned by the lower level for the current priorities; and (iii) the behavioral variables $F, F^{\text{ext}}, \{F_{ij}^{od}\}$, which arise as the optimality response of passenger choice to the offered network.

This is the formulation that the rest of the paper aims to solve. The lower-level optimality condition (57) is implicit and difficult to handle directly. The next section addresses this issue through a value-function penalty reformulation in the spirit of [34], [20], [21], and then presents the implemented solution algorithm together with the computational experiments performed on Spanish synthetic networks of increasing size and on a realistic Madrid-centred international instance.

V. PENALTY REFORMULATION, SOLUTION ALGORITHM, AND COMPUTATIONAL EXPERIMENTS

The bilevel problem (56)–(57) cannot be solved as it stands, because the lower-level optimality condition is implicit and the upper-level feasible set is therefore defined through the solution of another optimization problem. The strategy adopted here is to remove that implicit definition by means of a value-function penalty, in the spirit of the exact penalization results of [34] and the recent first-order penalty algorithms of [20], [21]. Concretely, recall that the lower-level value function associated with (26) is $v(\hat{\alpha}, \hat{\beta}) = \min_{Y \in \mathcal{X}_c} \Phi_c(Y; \hat{\alpha}, \hat{\beta})$. Bilevel feasibility (57) is then equivalent to the value-function equality

$$\Phi_c(X; \hat{\alpha}, \hat{\beta}) - v(\hat{\alpha}, \hat{\beta}) = 0, \quad (58)$$

because the lower-level objective Φ_c evaluated at the design X coincides with the optimal lower-level value if and only if X is itself a lower-level optimum. Replacing the implicit constraint (57) by the explicit equality (58) is exact, but it does not yet remove the difficulty, because v is itself the value function of an optimization problem and is therefore generally nonsmooth. The penalty reformulation we adopt instead promotes (58) to a

soft constraint by adding it to the upper-level objective with a positive weight $\gamma > 0$, which yields the single-level surrogate

$$\min_{(\hat{\alpha}, \hat{\beta}) \in \mathcal{H}, X \in \mathcal{X}_c} \Psi_\gamma(X, \hat{\alpha}, \hat{\beta}) := -\Pi(X) + \gamma(\Phi_c(X; \hat{\alpha}, \hat{\beta}) - v(\hat{\alpha}, \hat{\beta})). \quad (59)$$

Two facts are worth emphasizing. First, the penalized problem (59) is a single-level problem in $(X, \hat{\alpha}, \hat{\beta})$, whose feasible set is the simple Cartesian product $\mathcal{X}_c \times \mathcal{H}$ and whose objective combines the negative profit with a penalty on the distance to lower-level optimality. Second, for small values of γ the penalized problem prioritizes the upper-level profit term $-\Pi(X)$, allowing X to drift away from lower-level optimality if doing so increases profit; for large values of γ the penalty dominates and pushes X toward the lower-level optimal set, recovering the bilevel feasibility condition asymptotically. The classical value-function penalty results [34], [20] guarantee that, under mild regularity assumptions, stationary points of (59) approach neighborhoods of bilevel-stationary points of (56)–(57) as γ grows.

The implemented algorithm exploits this single-level penalty viewpoint while keeping the sparsity-inducing reweighting of fixed costs consistent with the current design. Two coupled iterations are needed, an inner one for the penalty and the market coefficients and an outer one for the reweighted ℓ_1 deployment term, because the deployment term (53) depends on the reference capacities $(\bar{S}, \bar{S}^h)$ that should themselves track the design returned by the algorithm. Concretely, at every outer iteration the reference capacities are written to the data files of the GAMS implementation [18], the inner loop alternates a pure lower-level solve and a penalized solve, the market coefficients are updated by projected steps using the gap between both solves, and only at the end of the inner loop the reference capacities are refreshed with the latest lower-level solution. The pure lower-level solve corresponds to a direct evaluation of $\arg \min_{Y \in \mathcal{X}_c} \Phi_c(Y; \hat{\alpha}, \hat{\beta})$, which is a convex problem under fixed deployment weights and is solved through the model `cvx-ll.gms`; the penalized solve corresponds to a single-level minimization of Ψ_γ at the current $(\hat{\alpha}, \hat{\beta})$, which is implemented in `cvx-sl.gms` with the profit term added with the opposite sign and the lower-level objective coupled through γ . Both models are solved with MOSEK as the underlying conic solver. The market-coefficient update at iteration (k, t) takes the form

$$\hat{\alpha}^{od, (k, t+1)} = \Pi_{[\underline{\alpha}, \bar{\alpha}]} \left(\hat{\alpha}^{od, (k, t)} - \mu_\alpha g_\alpha^{od, (k, t)} \right), \quad (60)$$

$$\hat{\beta}^{od, (k, t+1)} = \Pi_{[\underline{\beta}, \bar{\beta}]} \left(\hat{\beta}^{od, (k, t)} - \mu_\beta g_\beta^{od, (k, t)} \right), \quad (61)$$

where $\Pi_{[\cdot, \cdot]}$ denotes Euclidean projection onto the admissible interval, $\mu_\alpha, \mu_\beta > 0$ are step sizes, and the search directions $g_\alpha^{od}, g_\beta^{od}$ are the per-market gradient-like corrections computed from the discrepancy between the lower-level and the penalized contributions of market (o, d) to Φ_c . Intuitively, if a market is underemphasized by the inner lower-level model relative to the profit-oriented penalized solve, its $\hat{\alpha}^{od}$ is reinforced; if it is overemphasized, it is reduced. The diagonal

Algorithm 1: Penalty-based bilevel algorithm with outer sparsity updates

Input: Initial market coefficients $(\hat{\alpha}, \hat{\beta}) \in \mathcal{H}$, initial reference capacities $\bar{A}, \bar{S}, \bar{S}^h$, step sizes μ_α, μ_β , penalty parameter γ , numbers of outer and inner iterations $K_{\text{out}}, K_{\text{in}}$, tolerance $\eta > 0$.

```

1 for  $k = 1, \dots, K_{\text{out}}$  do
2   build the deployment term DEP in (53) from the
   current references  $\bar{A}, \bar{S}, \bar{S}^h$ ;
3   for  $t = 1, \dots, K_{\text{in}}$  do
4     solve the lower-level model
        $\min_{Y \in \mathcal{X}_c} \Phi_c(Y; \hat{\alpha}, \hat{\beta})$  via cvx-ll.gms and
       recover  $X^{\text{ll}}$ ;
5     evaluate the per-market lower-level
       contributions of  $\Phi_c$  at  $X^{\text{ll}}$ ;
6     solve the penalized model  $\min \Psi_\gamma(X, \hat{\alpha}, \hat{\beta})$  via
       cvx-sl.gms and recover  $X^{\text{pen}}$ ;
7     compute the projected update directions
        $g_\alpha^{\text{od}}, g_\beta^{\text{od}}$  from the difference between the
       lower-level and penalized market
       contributions;
8     update  $(\hat{\alpha}, \hat{\beta})$  via (60)–(61);
9     if  $|\Pi(X^{\text{pen},(t)}) - \Pi(X^{\text{pen},(t-1)})| / |\Pi(X^{\text{pen},(t-1)})| < \eta$  then
10      | break;
11   end
12 end
13 set  $\bar{A}, \bar{S}, \bar{S}^h \leftarrow A^{\text{ll}}, S^{\text{ll}}, S^{h,\text{ll}}$ ;
14 end

Output: Sparse capacities  $S^*, S^{h,*}, A^*$ , market shares
 $F^*, F^{\text{ext},*}$ , routing pattern  $\{F_{ij}^{\text{od},*}\}$ , and
calibrated market coefficients  $(\hat{\alpha}^*, \hat{\beta}^*)$ .

```

entries (o, o) are reset to their nominal values because self-markets do not contribute to the network design.

The full procedure is summarized in Algorithm 1. The outer loop is a majorization-minimization scheme for the deployment term: at every outer iteration, the reference capacities $(\bar{A}, \bar{S}, \bar{S}^h)$ are used to define the convex weighted penalty in (53), and at the end of the outer iteration they are refreshed with the latest lower-level solution. The inner loop is the value-function penalty algorithm: at every inner iteration, both the lower-level and the penalized models are solved, and the market coefficients are updated by projected steps. The choice of transferring the lower-level solution to the outer loop, rather than the penalized one, is deliberate, because the outer loop is meant to update the sparse approximation of the inner problem and the lower-level design is the one that approximates the value function $v(\hat{\alpha}, \hat{\beta})$. The early-stopping criterion of the inner loop is a relative-variation test on $\Pi(X)$ that prevents unnecessary iterations once the market coefficients have stabilized.

The whole scheme should be read as a nested successive-

approximation algorithm. The outer MM loop progressively improves the convex surrogate of the ℓ_0 activation cost by reweighting the linear penalties on S_i, S_i^h around the latest design, and the inner penalty loop progressively improves the alignment between the lower-level model and the airline profit objective by recalibrating the market coefficients $(\hat{\alpha}, \hat{\beta})$. Their combined action is what makes the approach computationally viable on instances that are well beyond the reach of the mixed-integer baseline.

a) Computational experiments.: The methodology is validated on four progressively larger Spanish-airline instances and on a realistic Madrid-centred international instance with Iberia-style coverage. The smallest instances are designed to allow a head-to-head comparison against a state-of-the-art commercial branch-and-bound solver, whereas the largest one is designed to illustrate the scalability of the proposed scheme on a setting where the mixed-integer baseline is no longer tractable.

The first instance is a 4-node Spanish network (4node_spain/) with airports MAD, BCN, PMI and AGP and twelve directed flight legs, used as a sanity check on a setting in which the MIP can still be solved to optimality within minutes. The second instance is a 6-node Spanish network (6node_spain/) that adds ALC and LPA to the previous airport set, generating thirty directed flight legs and roughly three dozen origin-destination markets. The third instance is an 8-node Spanish network (8node_spain/) that further adds TFS and IBZ, producing fifty-six directed legs and substantially denser interaction patterns. All Spanish instances share the same demand source, namely the public matrix CO261 of inter-airport passenger flows in Spain, and use the official YieldPKT data of the Spanish Ministry of Transport for fares; the underlying distance matrix is reconstructed from the official airport coordinates. Finally, the largest instance is the Madrid-centred international network (red_iberia_MAD/), which contains twenty-five candidate airports including the major Iberia destinations in Europe, North and South America (MAD, BCN, AGP, AMS, BIO, BRU, EZE, FCO, FRA, GRU, IBZ, JFK, LHR, LIM, LIS, LPA, MAH, MEX, MUC, MXP, ORY, PMI, SVQ, TFN, VLC) and the corresponding international demand and fare data.

In all four instances the proposed bilevel-sparsity scheme is implemented in Python coupled with the GAMS models `cvx-ll.gms` and `cvx-sl.gms`, and the resulting convex subproblems are solved with MOSEK. The mixed-integer baseline is the model of Section III solved with a commercial MILP solver under the same input data and a time limit of two hours per instance. For every Spanish instance, we report the mean wall-clock solution time, the number of activated airports and hubs, and the relative objective gap $|\Pi^{\text{BL}} - \Pi^{\text{MIP}}| / |\Pi^{\text{MIP}}|$ obtained when both designs are evaluated under the original mixed-integer profit (8); for the international Iberia instance, only the bilevel scheme produces solutions, since the mixed-integer baseline does not return a feasible incumbent within the time limit even for the

smallest budget regime considered. The aggregated comparison files `compare_bilevel_vs_mip_6node.csv` (and the corresponding 4-node and 8-node versions) collect these metrics for a sweep over budgets, hub-cost weights λ , market-coefficient initializations and step-size choices.

The qualitative findings are consistent across instance sizes. On the 4-node instance, the bilevel scheme matches the mixed-integer optimum on most budget regimes and runs in fractions of a second once the GAMS subproblems are warmed up, whereas the MIP requires tens of seconds to a few minutes. On the 6-node and 8-node instances the bilevel scheme remains in the second-to-minute regime, while the MIP frequently approaches the two-hour time limit; the resulting objective gaps depend strongly on the budget and on the choice of market-coefficient step sizes, with several configurations achieving small relative gaps and others showing the typical penalty-method behavior of trading optimality for runtime. The hub structure recovered by the bilevel scheme is qualitatively coherent with the one selected by the MIP, with MAD acting as the dominant hub in all instances and with secondary hubs appearing only when the budget is sufficient to support them. On the international Iberia instance, the bilevel scheme returns sparse network designs with a small number of European hubs (typically MAD and one of LHR, FRA or AMS) and a coherent allocation of long-haul services to the Americas; representative network maps are stored as PNG outputs in `red_iberia_MAD/plots_blo/`.

The overall picture that emerges from the experiments is twofold. On the one hand, the proposed continuous bilevel scheme is several orders of magnitude faster than the mixed-integer baseline on the small Spanish instances, and is the only one of the two that produces network designs at all on the realistic Madrid-centred international instance. On the other hand, the resulting designs depend nontrivially on the bilevel hyperparameters, and a careful calibration of the step sizes μ_α, μ_β and of the penalty weight γ is needed to keep the airline profit close to the mixed-integer benchmark. Both observations are in line with the behavior of value-function penalty methods reported in the recent first-order bilevel optimization literature [20], [21] and motivate the lines of further work discussed in the conclusions.

VI. CONCLUSION

This paper has developed a continuous bilevel framework for airline hub-and-spoke network design. The starting point is the observation that classical mixed-integer formulations describe the design problem accurately, but become computationally expensive when fixed deployment costs, passenger choice, multicommodity routing and profit-oriented objectives are included simultaneously. To address this limitation, we replace the discrete activation structure by sparse continuous capacities and recover passenger market shares through an entropy-regularized formulation that induces logit-type behavior. The resulting lower-level problem preserves the main economic ingredients of the original airline design problem while

remaining substantially more tractable than the corresponding mixed-integer model.

The second contribution is the bilevel calibration layer. Instead of fixing the relative importance of each origin-destination market exogenously, the proposed framework treats the market weights as upper-level decision variables. For a given set of weights, the lower level designs the network, installs capacities and estimates passenger flows; the upper level then adjusts those weights so that the induced design better matches the true objective of the airline, namely profit maximization. This interpretation is useful because the upper level does not directly impose a network topology. Rather, it learns how the continuous planning model should value each market in order to produce profitable network designs.

The fixed-cost component is handled through a reweighted ℓ_1 approximation derived from a logarithmic sparsity surrogate. This mechanism allows the continuous model to mimic build/no-build decisions without introducing binary variables, and it is updated through an outer majorization-minimization loop. The bilevel constraint is handled by a value-function penalty reformulation, which leads to an implementable algorithm based on two repeated convex solves: one for the lower-level design problem and one for the penalized profit-oriented problem. Together, these two ingredients provide a practical way of combining sparse network design and bilevel market calibration.

The computational experiments show that the proposed approach is especially attractive when the mixed-integer baseline becomes difficult or impossible to solve. On small Spanish instances, where the mixed-integer model can still return reliable solutions, the continuous bilevel method recovers very similar network designs and profit values while reducing solution times by several orders of magnitude. On larger instances, including the Madrid-centred international network, the mixed-integer formulation does not return useful feasible incumbents within the prescribed time limit, whereas the proposed method continues to produce sparse and economically coherent hub-and-spoke structures. These results suggest that the continuous bilevel approximation is not merely a computational shortcut, but a viable modeling alternative for larger airline network design problems.

Several limitations remain. The quality of the final design depends on the penalty parameter, the projected update steps used for the market weights, and the initialization of the sparsity loop. Although the experiments indicate that stable and profitable solutions can be obtained across a broad range of settings, a more systematic parameter-selection strategy would make the method easier to deploy in operational studies. In addition, the current numerical analysis focuses on aggregate profit, network topology and runtime. A richer assessment should also study passenger-level service quality, robustness to demand uncertainty, and the sensitivity of the selected hubs to fares, costs and budget assumptions.

Future work should therefore proceed in three directions. First, the penalty and reweighting parameters should be calibrated automatically, possibly through adaptive rules that react

to lower-level optimality gaps and changes in the selected network. Second, the model should be extended to include richer airline-planning features, such as frequency decisions, fleet heterogeneity, schedule connectivity and competition between multiple strategic carriers. Third, the continuous bilevel framework should be tested on larger and more detailed real-world datasets, where exact mixed-integer optimization is no longer a meaningful benchmark. These extensions would help determine how far continuous sparse bilevel optimization can be pushed as a practical tool for strategic airline network planning.

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